

Dillon Simeone

dillonsimeone@gmail.com

<https://dillonsimeone.github.io/Website/>

<https://github.com/DillonSimeone>

541-357-8716 (*Please use text or voice messages only*)

Professional Summary

Engineer specializing in creative coding, immersive technologies, and sound-reactive hardware. Experienced in full-cycle R&D of custom haptic devices, embedded systems, and audio-integrated systems. Passionate about leveraging deep technical skills to build innovative, accessible, and immersive experiences.

Work Experience

Lead Design Engineer | Universal Music Design (*July 2023 – August 2025*)

- **End-to-End Hardware Development:** Spearheaded the full lifecycle of haptic wearable devices, moving from conceptual sketches to functional prototypes. Designed custom PCBs and integrated ESP32 microcontrollers to process real-time audio data.
- **Embedded Systems & Firmware:** Developed low-latency firmware (C++/C) for sound-reactive systems, utilizing FFT (Fast Fourier Transform) algorithms to translate musical nuances into precise tactile and visual feedback.
- **Systems Integration:** Architected modular LED arrays and haptic driver circuits, ensuring seamless synchronization between audio inputs and sensory outputs for immersive accessibility tools.
- **Technical Leadership & Operations:** Established engineering protocols for inventory management (SnipeIT) and equipment procurement. Led prototype testing and validation to ensure reliability under live-performance conditions.
- **R&D Communication:** Authored technical research papers and grant proposals to secure funding, translating complex engineering concepts into accessible language for stakeholders and non-technical audiences.

Self-Employed Freelancer | Programmer & Web Designer (*May 2018 – July 2023*)

- **Full-Stack Development:** Created visually appealing and user-friendly websites for a diverse client base, including a Create-Read-Update-Destroy (CRUD) site using Gatsby and GraphQL to consume a WordPress RESTful API.
- **Performance & SEO:** Employed SEO best practices by optimizing for instant loading times, implementing critical CSS, and using cloud hosting services (Netlify/JamStack) to ensure robust and scalable solutions.
- **Problem-Solving:** Addressed complex technical issues to optimize website performance, security, and functionality.

Publications & Speaking

- Co-author: "GestoLumina: Gesture-Interpreted Light, Sound, and Haptics..." (2024). Aalborg University Press (VBN). <https://vbn.aau.dk/en/publications/gestolumina-gesture-interpreted-light-sound-and-haptics-towards-a/>
- "Sonic Agency: A Group Autoethnography of Technology-mediated Performance Practice by Deaf and Hard of Hearing Musicians" (2025). *ASSETS '25: Proceedings of the 27th International ACM SIGACCESS Conference on Computers and Accessibility*. <https://doi.org/10.1145/3663547.3746396>
- Guest Lecturer & Workshop Lead | Portland Community College (April 2025). Invited by faculty to design and lead a workshop on "Sonic Interface Design" for the MUC 262 (Digital/Analog Synthesis) class. <https://www.pcc.edu/harts/2025/04/21/sonic-interface-designer-dillon-simeone-visits-muc-262/>

Volunteer Work

Lead Technician / Systems Integrator | CymaSpace (May 2022 – December 2023)

- **Immersive Environment Design:** Directed the technical installation of large-scale, Meow Wolf-style art exhibits, integrating microelectronics, DMX lighting controls, and audio-reactive hardware.
- **Accessibility Engineering:** Collaborated with Deaf and Hard of Hearing teams to design inclusive media experiences, overcoming communication barriers through universal design principles.
- **Emerging Tech Integration:** Prototyped experimental AI-driven artistic displays, blending machine learning concepts with physical hardware to push the boundaries of interactive art.

Builder / Technician | FreeGeeks (March 2018 – May 2018)

- Diagnosed and repaired computer hardware and software, replacing faulty components (screens, RAM) and installing new operating systems. Averaged 6 computers processed per day.

Technical Skills

- **Programming:** C#, TypeScript, PHP, Java, Python, C++, Max MSP, Processing
- **Development:** HTML/CSS3/JavaScript, Android Studio, C# .NET
- **Frameworks:** REACT, Vue.js, Angular, Entity Framework, P5.JS
- **Database:** MySQL, NoSQL
- **Tools:** Microsoft Visual Studio, VS Code, Git, Github, Gitlab, Apache Subversion

Education

Portland Community College | Portland, OR *Associate of Applied Science, Creative Coding and Immersive Technologies* (June 2025)

- **Honors:** Highest Honors, President's List (all terms)
- Specialized in immersive audio, creative coding, and hardware design.

Rochester Institute of Technology | Rochester, NY *Web and Mobile Computing (Incomplete Coursework)*

- **Project Highlight:** Created a C# network application from scratch (no libraries) to scan a local network and establish an instant messaging room, demonstrating a strong understanding of the OSI model.
- **Project Highlight:** Built a 2-D action game in Java with a custom-animated engine for two-player keyboard input.
- **Project Highlight:** Developed a responsive Unix tutorial website with account creation, stats tracking, and a sandboxed Linux command environment.

Oregon State University | Corvallis, OR *General Engineering & Computer Science (Incomplete Coursework)*

- Completed foundational coursework in Computer Science I, General Chemistry, and Precalculus.

References

Academic & Research

- **Doga Buse Cavdir, PhD**
 - **Relationship:** Co-Author, *GestoLumina* Publications
 - **Contact:** doga.cavdir@gmail.com
 - **Credentials:** Postdoctoral Fellow, Aalborg University; PhD, Stanford (CCRMA)
- **Jesse Mejía**
 - **Relationship:** Faculty Department Co-Chair, Music and Sonic Arts
 - **Contact:** jesse.mejia15@pcc.edu
 - **Credentials:** Portland Community College
- **Ravi Kittappa**
 - **Relationship:** Instructor of Electrical Music
 - **Contact:** ravi.kittappa@pcc.edu
 - **Credentials:** Portland Community College